



A survey study of critical success factors in agile software projects in former Yugoslavia IT companies

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ABSTRACT

Determining the factors that have an influence on the success of the software development projects has been the focus of extensive research for more than 30 years. In recent years agile methodology of software development has become the dominant one for all kinds of software development projects. In this paper we present the results of empirical study for determining critical factors that influence the success of agile software projects which we conducted among senior developers and project managers from IT companies located in the former Yugoslavia countries within South Eastern Europe (SEE) region. This study is inspired by the similar study conducted 5 years ago (Chow and Cao, 2008). With this study we were not able to confirm the model developed in the previous study. Moreover it disconfirmed not only part of the factors, but very much questioned the whole scheme. However, we were able to shed additional light regarding agile software development in former Yugoslavia countries from SEE region as a reference region for investigating outsourced projects done in agile way.

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1. Introduction

A process of software development has been the focus of interest of many managers, engineers and researchers due to a large percentage of failures in the software industry. Failures range from inability to provide software solution that fits the requirements on time, to providing solutions that are a maintenance nightmare or in the worst case inability to provide any solution at all (abandoned software projects). One of the main problems that make software development so special and which is causing the above mentioned difficulties is that during the project both technology and the business environment change (Williams and Cockburn, 2003). That change is, due to technology advancement nowadays, even more dynamic than at the time Agile movement started its development; it is causing customers to have difficulties not only to state their needs in the beginning of the project but even to have a basic idea of what they need at that time and to form requirements only after a few iterations of the demo product. The process of forming requirements includes changes as well. The result of this situation is the development of a variety of

methodologies and practices that embrace changes like SCRUM, Extreme Programming, Lean Software Development, Kanban, Crystal, Feature Driven Development, etc.

Over the years agile methods have proven to overcome many of the problems stated above and have become dominant in the software industry. The agile approach is basically driven by self-organizing teams that have the power to coordinate their work on their own. This increases productivity, enables employees to learn, innovate, and finally makes them happy with what they do (Smite et al., 2010a). In the traditional plan-driven (waterfall) software development processes, work is coordinated by managers and there is a clear separation of roles (Moe et al., 2010). Similarly, in larger organizations there is a tendency of organizing people around component teams – grouping people in the way that they have the influence over the small part of the product thus giving the teams less control and losing the ability of close collaboration. Introducing new features in the organization of that kind requires synchronization of many different component teams (Larman, 2011). In the agile approach, a self-organizing team decides how work is coordinated and has the complete control over development process and introduction of new features. However, many organizations, and especially large organizations, still base their software development around plan driven or component teams. The transition of such teams to agile teams and how to overcome difficulties that occur during that process has

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been the subject of many case studies (Moe et al., 2010; Ogle et al., 2011). All of them agree that this kind of transition is a hard process. That process often has dead ends in means of trying and abandoning various software development practices that do not work for the current team. Sometimes it is filled with team members' frustration, skepticism, denying, but regardless of everything mentioned, achieved benefits ramify the cost of transition.

As Smite et al. (2010a) comprehend there is a growing need for companies to explore global sourcing leading to distributed software projects with geographically, temporally and socio-culturally dispersed teams facing additional challenges when trying to successfully implement agile values and principles. The research in Smite et al. (2010b) has shown that it is possible to successfully apply agile principles in distributed environment although these two can be considered as opposite extremes.

As stated by many authors (Smite et al., 2010b; Hansson et al., 2006) agile development practice has always been ahead of research. Academic research has mostly been involved with trying to understand what is going on and exploring in a scientific way techniques and procedures which were already established and used by the community of software developers and agile practitioners. One such example is work by Hansson et al. (2006) where they investigated the differences between industrial practices and agile development practices in several companies. They have concluded that the actual industrial practices used by companies were dependent on companies' characteristics and projects on which they worked on. Another study (Chow and Cao, 2008) among agile professionals has shown that only 10 out of 48 hypotheses were critical to success of agile projects. Based on the survey from that study in the work presented here we have tried to identify critical success factors in agile software projects on the sample of companies operating in SEE region (region sometimes referred as Western Balkans). Similarly to the authors of previous study we have used statistical methods to evaluate responses we received from our interviewees and verify obtained model. It can be debated whether model driven estimation based on statistics such as the one used in our study is the proper method to use when it comes to software engineering. For example, a study by Johansson (2000) suggests that the use of statistics is perhaps inappropriate in the field of software engineering due to all the difficulties associated with interpreting the results and many existing uncertainty factors. Similarly, Jørgensen and Boehm (2009) suggest that the use of both formal methods and expert judgment might be best. They also conclude that future efforts should be headed toward judgment based methods. Nevertheless, we have decided to use the same method as in previous study mainly to make more accurate comparison with the results of previous study. Since our study was not able to confirm the model developed in the previous study, in conclusions of this paper as part of our future research efforts we announce the use of AHP (Analytic Hierarchy Process) in order to create a better model.

Our survey was conducted among developers from over 20 companies. The general characteristic of our interviewees is that they mostly work in distributed environments meaning that they face additional challenges mentioned in the section above. We were interested in whether these specific environmental factors will influence the conclusions made by Chow and Cao (2008). In an article by Freudenberg and Sharp (2010) which was created as a result of panel discussion where practitioners identified the list of issues related to agile software development they would like to be researched, three of the top 10 items focus on distributed teams. This survey will hopefully show the difference to the previous study which can be attributed to both demographic and distributed teams effect.

2. Background (intro to agile methods)

Our survey showed that the most popular agile methods used among former Yugoslavia agile teams are: XP, Scrum, and Feature Driven Development. We have also received responses in which some hybrid methods were used. Interestingly, there were no reports on use of Lean or Kanban which are becoming increasingly popular in IT industry in recent years (Anderson, 2010; Kniberg and Skarin, 2010). The reason for this could be the delay in implementing the latest practices that exist in developers' communities from the more developed regions (for example in Silicon Valley). The possible cause of that delay could be attributed to the absence of modern trainings lead by qualified and experienced professionals that are available in more developed regions, or the fact that cutting edge projects are less likely to be outsourced to overseas teams. Nevertheless the most popular agile practices are there and were considered in our study.

Many agile methodologies share a lot of practices and have common characteristics and approach to projects, but each is unique in its own strategy of their implementation. In the following sections we will briefly describe each of the popular methods which we identified as being used by surveyed companies.

2.1. Extreme programming

Extreme Programming (XP) is one of several popular agile processes. It was originally described by Kent Beck (one of the authors of Agile manifesto) and has been proven to be very successful at many companies of all different sizes and within various industries. The focus of this approach is on customer satisfaction so it empowers developers to be able to respond to changing customer requirements and to deliver high-quality software quickly and continuously. Working software is delivered to customer typically in intervals of 1–3 weeks. XP improves software projects by embracing communication, simplicity, feedback, respect, and courage. The original XP recipe contained 12 rules: Planning Game, Small Releases, Customer Acceptance Tests, Simple Design, Pair Programming, Test-Driven Development, Refactoring, Continuous Integration, Collective Code Ownership, Coding Standards, Metaphor, and Sustainable Pace.

Like in every agile process these rules are not written in stone and over time certain rules were modified, new rules appeared and for example in Shore and Warden (2008) authors make distinction between two versions of XP described by Beck (1999) and Beck and Andres (2004) and even introduce their own approach to XP which emerged from their practical experience.

2.2. Feature driven development (FDD)

FDD was introduced by Jeff De Luca in 1997 and later as a result of collaboration with Peter Coad first incarnations of FDD appeared (Coad et al., 1999; Palmer and Felsing, 2002). FDD is a model-driven, short-iteration process. It begins with establishing an overall model shape and identification of features. Features are then grouped in work packages. One work package can be finished within a single iteration and it actually represents working software with which customer can play with.

FDD designs the rest of the development process around feature delivery using the following eight practices:

1. Domain object modeling
2. Developing by feature
3. Component/class ownership
4. Feature teams
5. Inspections

6. Configuration management
7. Regular builds
8. Visibility of progress and results

When comparing FDD to XP it seems that XP is better suited for volatile projects where user requirements are obscure or change often (Khramtchenko, 2004). This is mainly because XP avoids any activities that are not immediately required for the current implementation stage. In the same analysis FDD is determined to be more scalable than XP because XP heavily relies on communication within a team which becomes harder as team grows. FDD is better suited for large teams and within corporate environment.

2.3. SCRUM

Scrum can be observed as a lightweight management framework rather than a full process or methodology. It has broad applicability for managing and controlling iterative and incremental projects of all types.

Ken Schwaber, Mike Beedle, Jeff Sutherland and others have contributed significantly to the evolution of Scrum over the last decade. Over the last couple of years in particular, Scrum has garnered increasing popularity in the software community due to its simplicity, proven productivity, and ability to act as a wrapper for various engineering practices promoted by other agile methodologies.

In a nutshell, Scrum in general involves the following (Kniberg and Skarin, 2010):

- Splitting the organization into small, cross-functional, self-organizing teams (Scrum does not provide complete, detailed descriptions of how everything is to be done and much is left up to the team itself to decide).
- Splitting the project (work to be done) into a list called product backlog containing small, concrete deliverables such as features, bugs, non-functional requirements, etc. The list is sorted by priority and the estimate of relative effort for each item is done.
- Splitting the time and forming iterations which are not longer than 1 month. Iterations are called sprints and they end up with working code that can be presented and delivered to the customer.
- Optimizing the release plan and updating priorities in collaboration with the customer, based on insights gained by inspecting the release from previous iteration
- Optimizing the process itself by having retrospections after each iteration

Our survey has shown that Scrum is among the most used practices in surveyed companies and also it has been proven to scale to multiple teams across very large organizations.

3. Critical success factor

In order to analyze the results of our survey and compare them with results obtained by Chow and Cao (2008) we have decided to use same method, namely Critical Success Factor which will be shortly described in the following section.

Critical success factor (CSF) was introduced by Rockart (1979) as a method to help managers determine what information is the most relevant to them in order to allow them to successfully attain their goals in increasingly complex world. The method received additional background in Bullen and Rockart (1981). In that paper authors explain that the method reflects the fact that all good managers, most often subconsciously, have their own CSF's they use in order to help them manage throughout their careers. The method they introduced helps to identify these factors and make them

Table 1

Failure factors as identified by Chow and Cao (2008).

Dimension	Factor
Organizational	1. Lack of executive sponsorship
	2. Lack of management commitment
	3. Organizational culture too traditional
	4. Organizational culture too political
	5. Organizational size too large
People	6. Lack of agile logistical arrangements
	7. Lack of necessary skill-set
	8. Lack of project management competence
	9. Lack of team work
	10. Resistance from groups or individuals
Process	11. Bad customer relationship
	12. Ill-defined project scope
	13. Ill-defined project requirements
	14. Ill-defined project planning
	15. Lack of agile progress tracking mechanism
Technical	16. Lack of customer presence
	17. Ill-defined customer role
	18. Lack of complete set of correct agile practices
	19. Inappropriateness of technology and tools

public so managers can prioritize tasks better and improve resource allocation. Chow and Cao (2008) give the more recent examples of CSF method usage in the context of project management and software development project area. In this study we use the same CSF identification method they used in order to confirm or refine their findings in specific environment of former Yugoslavia companies.

4. Success factors in agile software development projects

In our analysis we have used success and failure factors identified by Chow and Cao (2008) and displayed in Tables 1 and 2.

They have grouped failure factors in four categories: organizational, people, process, and technical. In addition to failure factors, success factors contain a group related to projects. Authors have identified 19 failure factors and 36 success factors and have used four different attributes to identify perceived level of overall project success. These four attributes are the following: Quality (i.e. delivering good product or project outcome), Scope (meeting all requirements and objectives), Time (delivering on time), and Cost (delivering within estimated cost and effort). Using these factors Chow and Cao (2008) have generated a survey which we used for our study as well. The complete survey used in both studies is available in Appendix A. In the above mentioned paper authors have used Cronbach's alpha method and component factor analysis with Varimax rotation which allowed them to consolidate failure and success factors to the number of 12 (see Table 3).

In this paper we will be using these consolidated factors which now act as hypotheses, each linking Agile project success factor to four dimensions used to measure the level of success of that project.

5. Data collection

Data was collected with the help of web survey that was distributed to target population consisting of managers and senior developers in former Yugoslavia IT companies. Survey consisted of four sections. The Section 1 was on demographic data and information about the project. The Section 2 was on success factors, and the Section 3 was on perception of success. The Section 4 was reserved for additional comments. To measure importance of success factors and perception of success, a 7-point Likert scale was used.

Survey period lasted one month and we have collected 23 complete responses to our survey. Respondents belong to 23 different companies located in 4 former Yugoslavia countries out of 5 former Yugoslavia countries belonging to SEE region in 9 different towns.

Table 2
Success factors as identified by Chow and Cao (2008).

Dimension	Factor
Organizational	1. Strong executive support
	2. Committed sponsor or manager
	3. Cooperative organizational culture instead of hierarchical
	4. Oral culture placing high value on face-to-face communication
	5. Organizations where agile methodology is universally accepted
	6. Collocation of the whole team
	7. Facility with proper agile-style work environment
People	8. Rewards system appropriate for agile
	9. Team members with high competence and expertise
	10. Team members with great motivation
	11. Managers knowledgeable in agile process
Process	12. Managers who have light-touch or adaptive management style
	13. Coherent, self-organizing teamwork
	14. Good customer relationship
	15. Following agile-oriented requirement management
	16. Following agile-oriented project management process
	17. Following agile-oriented configuration management process
	18. Strong communication focus with daily face-to-face meetings
Technical	19. Honoring regular working schedule–no overtime
	20. Strong customer commitment and presence
	21. Customer having full authority
	22. Well-defined coding standards up front
Project	23. Pursuing simple design
	24. Rigorous refactoring activities
	25. Right amount of documentation
	26. Regular delivery of software
	27. Delivering most important features first
	28. Correct integration testing
	29. Appropriate technical training to team
	30. Project nature being non-life-critical
	31. Project type being of variable scope with emergent requirement
	32. Projects with dynamic, accelerated schedule
	33. Projects with small team
	34. Projects with no multiple independent teams
	35. Projects with up-front cost evaluation done
	36. Projects with up-front risk analysis done

Table 3
Consolidated factors or hypotheses as identified by Chow and Cao (2008).

Dimension	Factor
Organizational	v1. Management commitment
	v2. Organizational environment
	v3. Team environment
People	v4. Team capability
Process	v5. Customer involvement
	v6. Project management process
Technical	v7. Project definition process
	v8. Agile software techniques
Project	v9. Delivery strategy
	v10. Project nature
	v11. Project type
	v12. Project schedule

Table 4
Survey respondents' distribution per country.

Country	#Respondents
Serbia	18
Bosnia and Hercegovina	2
Croatia	2
Macedonia	1

Table 5
Survey respondents' distribution per town.

City, Country	#Respondents
Nis, Serbia	14
Belgrade, Serbia	3
Paracin, Serbia	1
Banja Luka, BH	1
Sarajevo, BH	1
Zagreb, Croatia	1
Rijeka, Croatia	1
Skopje, FYRM	1

Table 4 displays breakdown of number of respondents for each country. Table 5 displays the number of respondents for each town.

It may seem that large concentration of number of respondents located in Nis and overall number of respondents from Serbia can influence the results, but authors believe that this should not be true. The reason is the fact that all respondents work in different companies and on completely dissimilar projects and this is valid even if we observe IT companies from a single town as it is shown in the table by Stankovic (2011) covering technologies used by more than 40 IT companies operating in Nis. More, in cases where respondents' companies actually have outsourcing partners they are located in various countries. The fluctuation of developers also could negatively impact the survey since they could share their experiences and influence each other toward a non-representative direction, but this effect is diminished since authors paid strong attention to survey people that are not mutually connected. Furthermore, in order to statistically test the differences between respondents from various towns we have used ANOVA. The obtained results have shown that there are no statistically significant differences between respondents from various towns except for the v11 variable so we can safely assume that the distribution of the respondents has not influenced the results. We should also mention that since we had only single respondent per town for Croatia and Bosnia and Herzegovina we have put all Croatian respondents to one group and also all Bosnian respondents to the other group.

6. Data analysis and results

This research is an extension to exploratory study of Chow and Cao (2008) and therefore we use the same method they used for data analysis, namely multiple regression analysis. By leveraging this method we will try to determine the relationship between multiple independent variables (success factors) and the dependent variable (agile project success) as well as to establish relative predictive importance of the independent variables in the similar manner as authors of the previous study did.

For multiple regression analysis we use the following equation:

$$Y(Q, S, T, C) = \beta_1 SF_1 + \beta_2 SF_2 + \beta_3 SF_3 + \beta_4 SF_4 + \dots + \beta_{12} SF_{12}$$

where Y is Agile Project Success dependent variable, Q is the Quality dimension, S is the Scope dimension, T is the Time dimension, C is the Cost dimension, β_i is the Partial regression coefficient for the i th Success Factor (SF).

In order to verify obtained results we have statistically analyzed answers to our survey. The first analysis we have done is related to reliability statistics. We have analyzed reliability for each questions group except for v10, v11, and v12 since they contained only 1 question per group. We have also analyzed reliability for each dimension – Organizational, People, Process, Technical, and Project. The results are shown in Table 6. Table 6 also contains the number of items (questions) each of the analyzed groups/dimensions is explained

Table 6
Reliability statistics.

Dimension	Factor	Cronbach's alpha		No. of items	
		Question group	Dimension	Question group	Dimension
Organizational	v1. Management commitment	0.380	0.652	2	12
	v2. Organizational environment	0.442		6	
	v3. Team environment	0.548		4	
People	v4. Team capability	0.766	0.794	5	8
	v5. Customer involvement	0.740		3	
Process	v6. Project management process	0.665	0.778	6	9
	v7. Project definition process	0.710		3	
Technical	v8. Agile software techniques	0.663	0.680	5	7
	v9. Delivery strategy	0.415		2	
Project	v10. Project nature	–	0.318	1	3
	v11. Project type	–		1	
	v12. Project schedule	–		1	

with. We have high reliability when Cronbach's Alpha is between 0.7 and 1.

The multiple regression analysis was done on full regression model where all 12 independent variables were present and the objective was to find the variables with top scores which can be considered as critical success factors. The analysis (Appendix C) has shown that if we pick 12 factors from the model they can explain 88.126% of measurement variance.

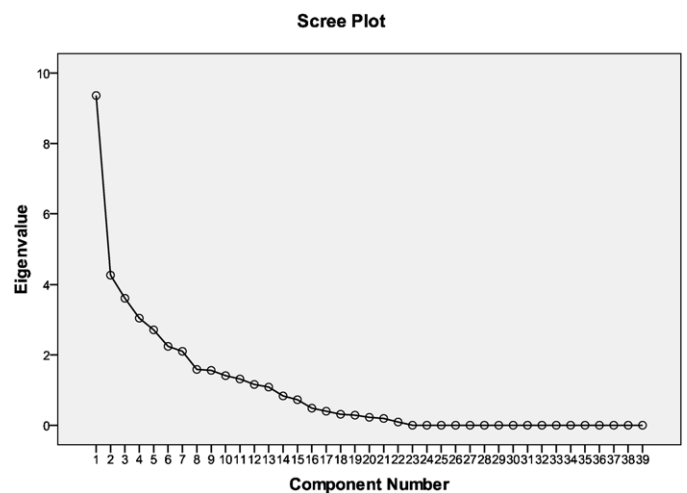
The structure of factors (which survey items are related to which factor) is determined based on the component matrix (Appendix D). Factors which we have obtained by using factor analysis are different than factors considered in study by Chow and Cao (2008). Therefore, we have made decision on how many factors to retain by using the Scree plot criteria (Fig. 1).

Scree plot does not give us clear answer on which factors to retain. Instead we have a couple of rules of thumb. For example one rule is to consider only those with eigenvalues over 1. We have instead plotted all the eigenvalues in their decreasing order (Fig. 1). The plot looks like the side of mountain, and “scree” refers to the debris fallen from a mountain and lying at its base. So the scree test proposes to stop analysis at the point where the mountain ends and the debris (error) begins. In our case this is somewhere between seventh and eighth factor.

The obtained component matrix shows which survey questions each factor contains. The first factor which we obtained by multiple regression analysis is related to 17 questions from the survey. 7 out of 17 questions that this factor is related to belong to the people dimension. 3 belong to organizational dimension, 3 to process and 4 are within technical dimension. For other factors we get questions distributions across many dimensions as well. The existence of factors which contain a variety of questions not closely related to each other in means of explaining a certain hypothesis does not allow us to generate meaningful explanations.

Table 7a
Quality prediction based on 12 factors.

Model		Unstandardized coefficients		Standardized coefficients	<i>t</i>	Sig.
		<i>B</i>	Std. error	Beta		
1	Constant	2.705	1.638		1.651	0.130
	v1	–0.165	0.210	–0.279	–0.786	0.450
	v2	0.184	0.380	0.201	0.484	0.639
	v3	0.241	0.204	0.390	1.182	0.265
	v4	–0.298	0.391	–0.418	–0.763	0.463
	v5	–0.038	0.165	–0.070	–0.228	0.824
	v6	0.695	0.478	0.864	1.454	0.177
	v7	0.072	0.180	0.127	0.400	0.698
	v8	–0.076	0.250	–0.115	–0.304	0.768
	v9	–0.048	0.242	–0.059	–0.196	0.848
	v10	–0.027	0.128	–0.063	–0.209	0.839
	v11	0.091	0.180	0.181	0.504	0.625
	v12	–0.051	0.204	–0.095	–0.251	0.807

**Fig. 1.** Scree plot criteria.

In the end we have decided to evaluate the model generated in the previous study. Factors we have used in analysis are the twelve consolidated factors from Table 3. The results we have obtained by analyzing the influence of these factors to the quality, scope, time and cost are given in Tables 7a–c, 8a–c, 9a–c, and 10a–c respectively. The displayed results help us to determine which of the 12 hypotheses is true for the given attribute that measures the agile project success.

Column B in Table 7a is filled with coefficients of the regression model used for predicting Q (quality) dimension based on twelve factors (reflected as hypotheses H1a–H12a from Table 11). By looking at the significance column (Sig) we can observe that none of the

Table 7b
Quality variance.

Model	R	R square	Adjusted R square	Std. error of the estimate
1	0.707	0.500	–0.100	0.788

Table 7c
Quality prediction ability.

Model		Sum of squares	df	Mean square	F	Sig.
1	Regression	6.219	12	0.518	0.834	0.623
	Residual	6.216	10	0.622		
	Total	12.435	22			

12 given predictors is having statistically significant impact on the final prediction (all values are higher than 0.05).

This model explains 50% of quality dimension variance (*R* square column in [Table 7b](#)).

Based on the obtained significance (column Sig in [Table 7c](#)) of $\text{sig} = 0.623$ which is greater than 0.05 we can conclude that this model does not have an important prediction ability.

[Table 8a](#) shows the coefficients of the regression model used for predicting *S* (scope) dimension based on H1b–H12b. Again, based on the Sig. column we can conclude that none of the 12 given predictors is having statistically significant impact on the final prediction. [Tables 8b and 8c](#) show that the model explains 58.9% of scope dimension variance, having significance of 0.395 and consequently does not have important prediction ability.

[Tables 9a–9c](#) similarly show the results of the regression model for predicting *T* (time) dimension based on the same factors as previous two models. Based on the Sig column in [Table 9a](#) we conclude that only H10c (Project nature) has significant influence ($\text{sig} = 0.029$) to whether the project will be finished on time. The model explains 65.2% of *T*'s variance ([Table 9b](#)) having significance of 0.244 which is still higher than 0.05. The model does not have

important prediction ability. [Tables 10a–10c](#) show the results of the regression model for predicting *C* (cost) dimension based on the same 12 factors from previous models. As we can observe from the Sig column in [Table 10a](#), predictors H6d (Project management process), H7d (Project definition process), H10d (Project nature), and H12d (Project schedule) have statistically important influence on project cost. The model explains 81.6% of *C*'s variance ([Table 10b](#)). This is the only model that has significant prediction ability with significance of $\text{sig} = 0.023$ (less than 0.05).

6.1. Summary of hypothesis testing results

[Table 11](#) outlines the results we obtained in our study. The results from the former study are given as well so we can observe whether CSF's obtained with our study match with CSF's obtained with previous study. Unfortunately, the only match between these two studies we can identify is related to the absence of evidence that some assumed prerequisites for success of Agile projects such as strong executive support (v1), organizational environment (v2), or Agile-appropriate project types (v11) are actually critical factors for success.

Table 8a
Scope prediction based on 12 factors.

Model		Unstandardized coefficients		Standardized coefficients	<i>t</i>	Sig.
		<i>B</i>	Std. error	Beta		
1	Constant	4.230	1.455		2.909	0.016
	v1	0.021	0.187	0.037	0.114	0.911
	v2	0.574	0.337	0.643	1.704	0.119
	v3	–0.016	0.181	–0.026	–0.087	0.933
	v4	–0.626	0.347	–0.897	–1.804	0.101
	v5	–0.037	0.147	–0.070	–0.252	0.806
	v6	0.656	0.425	0.833	1.544	0.154
	v7	–0.069	0.160	–0.123	–0.430	0.677
	v8	0.182	0.222	0.281	0.819	0.432
	v9	–0.129	0.215	–0.165	–0.601	0.561
	v10	0.052	0.114	0.125	0.456	0.658
	v11	–0.169	0.159	–0.346	–1.061	0.313
	v12	–0.123	0.181	–0.234	–0.679	0.513

Table 8b
Scope variance.

Model	R	R square	Adjusted R square	Std. error of the estimate
1	0.767	0.589	0.095	0.700

Table 8c
Scope prediction ability.

Model		Sum of squares	Df	Mean square	F	Sig.
1	Regression	7.014	12	0.584	1.193	0.395
	Residual	4.900	10	0.490		
	Total	11.913	22			

Table 9a

Time prediction based on 12 factors.

Model		Unstandardized Coefficients		Standardized coefficients	<i>T</i>	Sig.
		<i>B</i>	Std. error	Beta		
1	Constant	−0.161	2.820		−0.057	0.955
	v1	−0.420	0.362	−0.343	−1.159	0.273
	v2	0.518	0.653	0.275	0.793	0.446
	v3	−0.127	0.351	−0.099	−0.360	0.726
	v4	−0.062	0.673	−0.042	−0.093	0.928
	v5	−0.018	0.285	−0.016	−0.063	0.951
	v6	1.071	0.823	0.645	1.300	0.223
	v7	0.126	0.311	0.107	0.406	0.693
	v8	−0.305	0.431	−0.223	−0.708	0.495
	v9	−0.088	0.417	−0.053	−0.211	0.837
	v10	0.563	0.221	0.645	2.554	0.029
	v11	0.090	0.309	0.088	0.292	0.776
	v12	−0.409	0.351	−0.370	−1.167	0.270

Table 9b

Time variance.

Model	<i>R</i>	<i>R</i> square	Adjusted <i>R</i> square	Std. error of the estimate
1	0.808	0.652	0.235	1.357

Table 9c

Time prediction ability.

Model		Sum of squares	Df	Mean square	<i>F</i>	Sig.
1	Regression	34.541	12	2.878	1.563	0.244
	Residual	18.416	10	1.842		
	Total	52.957	22			

Table 10a

Cost prediction based on 12 factors.

Model		Unstandardized coefficients		Standardized coefficients	<i>t</i>	Sig.
		<i>B</i>	Std. error	Beta		
1	Constant	4.326	1.820		2.377	0.039
	v1	−0.423	0.234	−0.390	−1.811	0.100
	v2	0.119	0.422	0.071	0.281	0.784
	v3	−0.280	0.227	−0.247	−1.236	0.245
	v4	−0.607	0.434	−0.464	−1.397	0.193
	v5	0.060	0.184	0.061	0.326	0.751
	v6	1.199	0.531	0.813	2.257	0.048
	v7	0.488	0.200	0.468	2.435	0.035
	v8	−0.027	0.278	−0.022	−0.097	0.925
	v9	−0.314	0.269	−0.214	−1.165	0.271
	v10	0.402	0.142	0.518	2.821	0.018
	v11	0.022	0.199	0.024	0.108	0.916
	v12	−0.514	0.226	−0.523	−2.270	0.047

Table 10b

Cost variance.

Model	<i>R</i>	<i>R</i> square	Adjusted <i>R</i> square	Std. error of the estimate
1	0.903	0.816	0.596	0.876

Table 10c

Cost prediction ability.

Model		Sum of squares	Df	Mean square	<i>F</i>	Sig.
1	Regression	34.069	12	2.839	3.701	0.023
	Residual	7.670	10	0.767		
	Total	41.739	22			

Table 11

Summary of hypothesis testing results (“√ x” – hypothesis proven by Chow and Cao (2008) and not confirmed in our study; “x √” – hypothesis not proven in previous study but proven here).

	Quality	Scope	Timeliness	Cost
1. Management commitment	H1a	H1b	H1c	H1d
2. Organizational environment	H2a	H2b	H2c	H2d
3. Team environment	H3a √ x	H3b	H3c	H3d
4. Team capability	H4a	H4b	H4c √ x	H4d √ x
5. Customer involvement	H5a	H5b √ x	H5c	H5d
6. Project management process	H6a √ x	H6b	H6c	H6d x √
7. Project definition process	H7a	H7b	H7c	H7d x √
8. Agile software engineering	H8a √ x	H8b √ x	H8c	H8d
9. Delivery strategy	H9a	H9b √ x	H9c √ x	H9d √ x
10. Project nature	H10a	H10b	H10c x √	H10d x √
12. Project type	H11a	H11b	H11c	H11d
13. Project schedule	H12a	H12b	H12c	H12d x √

Please note: Double checkmark symbol “√ √” is not present in any case (it would have appeared in the case of the hypothesis proven in both studies).

In order to further investigate why the results from the two studies are different we have statistically analyzed the differences in responses in relation to the agile method used (Scrum, XP, FDD), internal company organization (outsourcing, has own product, mixed (distributed) – internal product is a part of an outsourcing project), and the respondent position (Intermediate, Senior, Team Leader). Since the majority of respondents were using Scrum, the possibility that this will mediate the obtained CSF's was investigated as well. By using ANOVA we have confirmed that there are no significant differences between respondents using Scrum and any other method. Also significant difference was not found among respondents working at various positions within companies.

However, significant difference was found between respondents working in outsourcing companies or companies having own product and respondents working in ‘mixed’ companies. This difference can be explained with the factor of distributed environment since a team in ‘mixed’ case typically consists of two subparts with people that are not collocated. This is somewhat different than the typical outsourcing project where a team can be entirely collocated in the outsourcing country and where the only distributed relation is the relation between customer or a project lead and the team itself. Mixed type companies deserve further explanation since a number of respondents we have surveyed works in IT companies of that kind. A typical example of the team organization in this case can be described as following: a team consisting of a part located in e.g. Serbia and a part located in e.g. USA work on the same project. The team in Serbia uses internal software product (for example search engine) developed to boost performance or for further reuse in similar projects. Serbian team integrates that internal product in the project on which they work together with USA developers who do not have prior experience with the internal product of Serbian team. This puts additional new challenges which are not discussed in this paper. Rather, by using statistics we have identified their existence and influence to our results. Furthermore we have identified the actual factors which are influenced by this setup. The identified factors are: customer involvement (v5), project management process (v6), agile software engineering (v8) and project nature (v10). Interestingly, 3 out of 5 hypotheses we have proved in our study and 4 out of 10 proved in the previous study are related to these factors. By using Fischer's LSD post hoc method we have confirmed that for the identified factors the difference in results exists between ‘mixed’ and own product groups and ‘mixed’ and outsourcing groups and not between outsourcing and own product groups. This leads us to conclusion that the most probable reason for the difference in obtained results between our and previous study is the distributed environment of ‘mixed’ companies.

What is also interesting is the fact that 4 out of 5 hypotheses which proved right belong to the cost domain. One obvious reason

for this is that the cost is an important driving factor when deciding to outsource projects to SEE countries. The developers and managers from former Yugoslavia are aware of that fact which further causes to pay more attention on it. Our study has successfully identified that situation.

6.2. Research limitations

As we have already mentioned the conducted research has certain limitations. The first is that only three out of a seven agile methods mentioned in previous study were used on projects described by our respondents. The second is strong bias toward Scrum which can be considered as an evolution to the bias reported in previous study toward XP. Scrum is nowadays reported to be the most widely accepted agile practice (State of Agile Survey, 2011). Our study has shown that IT companies from former Yugoslavia countries are not the exception. By using the invitation based survey instead of publicly available we have avoided the bias that certain respondents could have toward representing better project success than what it actually was. We did not pick the survey participants among people who have direct benefit of subjective representation of project success. However, the research setting in our and in the previous study did not address how well agile methods have been followed by the respondents. Perhaps better and more accurate model can be obtained if we extract only respondents who were properly following agile methods they employ for software development. We see this as a potential direction to follow in our future research related to the subject. The final and possibly the most important limitation is the sample size. Although sample size of 23 is lot smaller than in the previous study, the relative ratio of respondents to the overall number of practitioners in the geographical area covered by our survey is better than the same ratio from the previous study (109 respondents from all over the world versus 23 for the former Yugoslavia part of SEE region only).

7. Conclusions

In this study we have tried to verify the classification of critical success factors previously described in study by Chow and Cao (2008). The regression analysis done on our collected data has introduced three more factors that could potentially be considered as critical success factors in terms of Timeliness and Cost (H10 in terms of Timeliness and Cost, and H7 together with H12 in terms of Cost). On the opposite side, the obtained results did not confirm that either of the factors from the previous study can be considered as critical success factor in the space of former Yugoslavia IT companies. However, our results match with the results from the previous study in suggesting that strong executive support

and project type has no influence on the success of agile project. Our study similarly suggests that organization environment (agile-style work facility) is not a factor with strong influence on the success of agile projects carried out in former Yugoslavia IT companies. In our study only 5 out of 48 hypotheses proved right and together with 10 out of 48 proven in the previous study leads us to conclusion that we should try to formulate different success factors or try to model the success of agile projects with different method.

A recently published survey of research literature (McLeod and MacDonell, 2011) covering period 1996–2006 with focus on empirical analyses has proposed a new classification framework of the types of factors that are believed to have important influence on project success. Their classification largely overlaps with the classification observed in this study. However as we did not succeed to prove the critical connection of the large majority of described factors to project outcome and as we are aware of the flaws that empirical studies can have, such as shown with infamous “IROP study” by Zeller et al. (2011), we think that different method, perhaps AHP method (Analytic Hierarchy Process) combined with the latest classification from McLeod and MacDonell (2011) will perhaps reveal more relevant results.

Many research results are available to us that show the benefits agile practices have to the project cost and timeliness. As a possible other direction to consider in the near future we plan on investigating how we can expand the domain on which agile practices should be applied and consider their use to stop building worthless software at the first place (Patton, 2011).

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Appendix A. Survey instrument

Agile software development project survey

Section 1 – Demographics

Thank you very much for agreeing to spend a few minutes of your time to complete this survey. If you have been involved with more than one agile project, please pick one (either successful or failed) that was most relevant or most telling with regard to critical success factors of such a project.

Section 1.1

For questions 1–5 please provide some basic information regarding the agile project.

1. Project description (i.e. what the software was about):
2. Agile method used:
3. Size of the project (number of project team members):
4. Length of the project (in months):
5. Location of the project (country):

Section 1.2

For questions 6–13 please provide some basic information regarding your organization and yourself (all information provided will be kept completely confidential):

6. Company name (optional):
7. Company size (ranges of number of employees):
8. Company revenues (ranges of annual sales dollar amounts):
9. Company industry (selection of pre-determined industries):

10. Your job responsibility in the project (project manager, team lead, team member, customer, organization management, other)
11. Your level of experience with agile projects (in years):
12. Number of agile project you have been involved with:
13. Please provide your name, address, phone and email information, should we need to get some clarification regarding your response to this survey (optional):

Section 2 – Success factors of the agile project

This section includes all the possible success factors of software development projects using agile methods, which had been compiled and consolidated from the academic and professional literature. Responses to each of the following statements range from 1 to 7 as follows:

- 1 Strongly disagree
- 2 Disagree
- 3 Somewhat disagree
- 4 Neither disagree or agree
- 5 Somewhat agree
- 6 Agree
- 7 Strongly agreeN/A–Not applicable/Don't know

Section 2.1 – Organizational dimension

14. The project received strong executive support. “Executive” may mean the whole Board of Directors or the CEO, CFO, CIO, etc. who influenced the decision-making:
15. The project had a committed sponsor or a committed organization manager. An example of a committed sponsor/manager would be one who would stand up to critics and vouch for the agile method in a non-agile organizational environment:
16. The organization had a cooperative culture instead of hierarchal. A cooperative culture is one that fosters ad-hoc teams driven by the needs of the job at hand (e.g. start-up organizations) while a hierarchal culture is one that has clear divisions of responsibility and authority (e.g. established, large organizations):
17. The organization had an oral culture placing high value on fluid, face-to-face communication style:
18. Agile methodology was universally accepted in the organization:
19. The organization had a reward system that was appropriate for agile behavior. An example of such a reward system would be one that recognizes both individual and team contributions, and that rewards results of the agile pilot projects:
20. The project team was collocated, i.e. all team members worked in the same location for ease of communication and casual, constant contact:
21. The project team worked in a facility with proper agile-style work environment, e.g. a dedicated office with pair programming workstations, communal area, ample wall spaces for postings, no separate cubicles or offices, etc.:

Section 2.2 – People dimension

22. The selected project team members had high technical competence and expertise (problem solving, programming, subject matter):
23. Project team members had great motivation and were committed to the project success:
24. Project management was knowledgeable in agile principles and processes:
25. Project management had light-touch and/or adaptive management style, e.g. encouraging creative, flexible working

environment while taking advantage of mutual interactions among the project's various parts and steering them toward continuous learning and adaptation:

26. The project team worked in a coherent, self-organizing team-work manner, i.e. relying on the collective ability of an autonomous team to solve problems and adapt to changing conditions:
27. Project management had a good relationship with the customer:

Section 2.3 – Process dimension

28. The project scope and objectives were well-defined:
29. The project followed agile-oriented requirement process, e.g. specifying initial requirements at a very high level, leaving much room for interpretation and adaptation as the project progressed:
30. The project followed agile project management style, e.g. plans generally not being documented in great detail, and deviations and changes being readily accepted and incorporated into the project plan:
31. The project followed agile-oriented configuration management process, e.g. employing good version control or source code management to accommodate the refactoring efforts and frequent builds:
32. The project manager followed an agile-friendly progress tracking mechanism, e.g. using flexible time-boxing or rapid-pace progress measurement techniques instead of document milestones or work breakdown structure:
33. The project had strong communication focus and rigorous communication schedule, i.e. face-to-face and instant communication channels (between team members, between team and management, and between team and customers), daily stand-up meetings, build cycle meetings, etc.:
34. The project honored regular working schedule, i.e. 40-hour work week, no overtime:
35. The project had strong customer commitment and presence, i.e. having at least one customer representative on site working hard and full-time as a member of the project team:
36. The customer representative on the project had full authority and knowledge to make decisions on-site, such as approving, disapproving, and prioritizing project requirements and changes:

Section 2.4 – Technical dimension

37. The project imposed a well-defined coding standards up front:
38. The project pursued simple design, e.g. programmers used the simplest possible design for each module to avoid waste and to facilitate cooperative work:
39. The project pursued vigorous refactoring activities to ensure the results are optimal and to accommodate well all changes in requirements:
40. The project maintained right amount of documentation for agile purpose, i.e. not too focused on producing elaborate documentation as milestones but not ignoring documentation altogether either:

41. The project followed continuous and rigorous unit and integration testing strategy for each and every iteration:
42. The project delivered working software regularly within short periods of time:
43. The project delivered most important features first:
44. The project employed proper platforms, technologies, and tools suitable for agile practice, e.g. object-oriented development techniques, tools supporting rapid iterative development, processes supporting refactoring, etc.:
45. The project provided appropriate technical training to team, including training on subject matter and agile processes:

Section 2.5 – Project dimension

46. The project nature was a non-life-critical software project, although it could be a business mission-critical software. (Examples of life-critical projects are certain advanced weapons programs or air traffic control programs):
47. The project type was of variable scope with emergent requirements:
48. The project had a dynamic, accelerated schedule:
49. The project had a small team size (20 members or less):
50. The project had no multiple, independent teams working together:
51. The project had up-front, detailed cost evaluation done and approved:
52. The project had up-front risk analysis done and evaluated for using agile method:

Section 3 – Perception of success of the agile project

This section includes aspects of your perceived level of success of the agile software development project at hand. Responses to each of the following statements range from 1 to 7 as follows:

- 1 Very unsuccessful
- 2 Unsuccessful
- 3 Somewhat unsuccessful
- 4 Neutral
- 5 Somewhat successful
- 6 Successful
- 7 Very successful

53. The project was successful in terms of quality of the project outcome or of the resulting software product:
54. The project was successful in terms of scope and requirements of the project being met:
55. The project was successful in terms of timeliness of project completion:
56. The project was successful in terms of costs and efforts being under budget or within estimates:

Section 4 – Additional comments

This section includes one free-form text area where you are invited to enter any additional comments on any matter which has not been covered in the survey. Your input may be used for follow-up for clarification or for further exploration if necessary:

57. Please enter any additional comments or thoughts here:

Appendix B. Mapping variables to survey questions

Variable	Item attribute	Variable type	Data type	Survey question
Demographic profile	Thirteen different attributes on personal, organizational, and project information		Nominal or ratio	1–13
v1. Management commitment	i1. Strong executive support	Independent	Interval	14
	i2. Committed sponsor or manager			15
v2. Organization environment	i3. Cooperative organizational culture instead of hierarchal	Independent	Interval	16
	i4. Oral culture placing high value on face-to-face communication			17
	i5. Organizations where agile methodology is universally accepted			18
	i6. Reward system appropriate for agile			19
	i8. Facility with proper agile-style work environment			21
	i31. Appropriate platform technologies and tools			44
v3. Team environment	i7. Collocation of the whole team	Independent	Interval	20
	i13. Coherent, self-organizing teamwork			26
	i36. Projects with small team			49
	i37. Projects with no multiple independent teams			50
v4. Team capability	i9. Team members with high competence and expertise	Independent	Interval	22
	i10. Team members with great motivation			23
	i11. Managers knowledgeable in agile			24
	i12. Managers who have light-touch or adaptive management style			25
	i32. Appropriate technical training to team			45
v5. Customer involvement	i14. Good customer relationship	Independent	Interval	27
	i22. Strong customer commitment and presence			35
	i23. Customer having full authority			36
v6. Project management process	i16. Following agile-oriented requirement management process	Independent	Interval	29
	i17. Following agile-oriented project management process			30
	i18. Following agile-oriented configuration management process			31
	i19. Good progress tracking mechanism			32
	i20. Strong communication focus with daily face-to-face meetings			33
	i21. Honoring regular working schedule			34
v7. Project definition process	i15. Project scope well-defined	Independent	Interval	28
	i38. Projects with up-front cost evaluation done			51
	i39. Projects with up-front risk analysis done			52
v8. Agile software engineering techniques	i24. Well-defined coding standards up front	Independent	Interval	37
	i25. Pursuing simple design			38
	i26. Rigorous refactoring activities			39
	i27. Right amount of documentation			40
	i28. Correct integration testing			41
v9. Delivery strategy	i29. Regular delivery of software	Independent	Interval	42
	i30. Delivering most important features first			43
v10. Project nature	i33. Project nature being non-life-critical	Independent	Interval	44
v11. Project type	i34. Project type being of variable scope with emergent requirement	Independent	Interval	45
v12. Project schedule	i35. Projects with dynamic, accelerated schedule	Independent	Interval	46

Appendix C. Factor analysis

Component	Initial eigenvalues			Extraction sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
1	9.362	24.005	24.005	9.362	24.005	24.005
2	4.267	10.942	34.947	4.267	10.942	34.947
3	3.610	9.257	44.204	3.610	9.257	44.204
4	3.044	7.805	52.010	3.044	7.805	52.010
5	2.712	6.953	58.963	2.712	6.953	58.963
6	2.243	5.751	64.714	2.243	5.751	64.714
7	2.101	5.388	70.102	2.101	5.388	70.102
8	1.587	4.069	74.171	1.587	4.069	74.171
9	1.559	3.998	78.169	1.559	3.998	78.169
10	1.410	3.615	81.784	1.410	3.615	81.784
11	1.314	3.369	85.153	1.314	3.369	85.153
12	1.159	2.972	88.126	1.159	2.972	88.126
13	1.084	2.781	90.906			
14	0.833	2.135	93.042			
15	0.725	1.859	94.901			
16	0.485	1.244	96.145			
17	0.398	1.022	97.166			
18	0.311	0.798	97.965			
19	0.286	0.733	98.697			
20	0.225	0.576	99.273			
21	0.192	0.493	99.766			
22	0.091	0.234	100.000			
23	0.000	0.000	100.000			
24	0.000	0.000	100.000			
25	0.000	0.000	100.000			

Appendix C (Continued)

Component	Initial eigenvalues			Extraction sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
26	0.000	0.000	100.000			
27	0.000	0.000	100.000			
28	0.000	0.000	100.000			
29	0.000	0.000	100.000			
30	0.000	0.000	100.000			
31	0.000	0.000	100.000			
32	0.000	0.000	100.000			
33	0.000	0.000	100.000			
34	0.000	0.000	100.000			
35	0.000	0.000	100.000			
36	0.000	0.000	100.000			
37	0.000	0.000	100.000			
38	0.000	0.000	100.000			
39	0.000	0.000	100.000			

Appendix D. Component matrix

	Component						
	1	2	3	4	5	6	7
1							
p32	0.786	−0.187	−0.170	0.194	−0.306	0.047	−0.103
p15	0.760	−0.270	−0.047	0.094	−0.009	0.221	0.287
p26	0.757	−0.049	−0.006	−0.110	−0.062	−0.041	−0.504
p24	0.754	0.295	−0.219	0.044	−0.002	−0.322	−0.077
p10	0.745	−0.052	0.103	−0.064	−0.448	−0.032	−0.071
p11	0.712	0.095	0.018	−0.099	−0.156	−0.158	0.303
p21	0.711	−0.146	−0.144	0.199	−0.054	0.063	0.342
p23	0.672	0.115	−0.303	0.196	0.158	0.174	−0.122
p27	0.649	−0.043	0.288	−0.317	0.284	0.267	−0.286
p18	0.626	−0.151	−0.430	−0.511	0.031	−0.151	0.028
p8	0.604	0.443	−0.253	0.131	0.296	−0.139	−0.035
p4	0.536	−0.261	0.381	0.112	−0.069	−0.416	−0.037
p12	0.524	−0.469	0.314	−0.273	0.165	−0.184	0.108
p22	0.484	−0.280	−0.334	0.328	−0.125	0.233	0.065
p29	0.482	0.451	0.336	−0.264	−0.131	0.203	−0.016
p28	0.477	0.236	0.091	−0.089	−0.045	−0.078	−0.060
p14	0.453	−0.059	−0.093	0.119	0.265	0.381	−0.073
p16	0.425	−0.418	0.144	−0.148	0.022	0.386	0.197
p17	0.419	−0.253	0.274	−0.332	0.111	0.177	0.156
p7	0.412	0.289	0.261	0.373	0.288	−0.362	−0.027
2							
p19	0.518	− 0.648	−0.117	−0.360	0.127	−0.084	0.125
p25	0.167	− 0.615	0.115	0.233	0.036	−0.281	−0.134
p5	0.324	0.603	0.119	−0.421	0.024	−0.153	0.100
p20	0.432	0.555	−0.257	0.490	0.011	−0.041	−0.131
p34	−0.095	0.544	0.170	−0.369	0.510	0.104	−0.018
p6	0.433	0.533	0.085	−0.176	0.016	0.281	0.202
p35	0.175	0.379	−0.342	−0.342	0.342	0.242	−0.370
3							
p36	0.395	0.390	0.687	0.257	−0.172	−0.170	−0.027
p31	0.432	0.098	− 0.657	−0.298	−0.097	−0.034	−0.025
p33	0.090	0.331	0.587	0.244	−0.010	−0.060	0.239
p3	0.003	−0.369	0.552	−0.122	0.088	0.352	−0.204
p13	0.462	−0.018	0.521	0.208	0.044	−0.024	−0.482
4							
p38	0.091	−0.063	−0.368	0.666	0.354	0.098	0.089
p39	0.465	−0.193	−0.009	0.526	0.329	0.278	−0.142
5							
p9	0.251	0.228	−0.004	0.085	− 0.749	0.011	0.139
p37	0.017	−0.138	0.393	0.177	0.529	0.095	0.192
6							
p1	0.359	−0.210	0.009	−0.168	0.285	− 0.578	0.270
p30	0.167	0.163	0.228	0.076	−0.463	0.545	0.248
7							
p2	0.123	0.261	−0.021	0.068	0.380	0.051	0.745

Appendix E. Analysis of variance between town groups

		Sum of squares	df	Mean square	F	Sig.
v1	Between groups	10.361	5	2.072	0.827	0.548
	Within groups	42.595	17	2.506		
	Total	52.957	22			
v2	Between groups	83.512	5	16.702	0.773	0.582
	Within groups	367.357	17	21.609		
	Total	450.870	22			
v3	Between groups	33.224	5	6.645	0.508	0.766
	Within groups	222.429	17	13.084		
	Total	255.652	22			
v4	Between groups	87.122	5	17.424	0.597	0.703
	Within groups	496.095	17	29.182		
	Total	583.217	22			
v5	Between groups	176.122	5	35.224	2.660	0.059
	Within groups	225.095	17	13.241		
	Total	401.217	22			
v6	Between groups	82.389	5	16.478	0.472	0.792
	Within groups	593.524	17	34.913		
	Total	675.913	22			
v7	Between groups	36.969	5	7.394	0.559	0.730
	Within groups	224.857	17	13.227		
	Total	261.826	22			
v8	Between groups	102.389	5	20.478	0.522	0.757
	Within groups	667.524	17	39.266		
	Total	769.913	22			
v9	Between groups	30.361	5	6.072	0.627	0.681
	Within groups	164.595	17	9.682		
	Total	194.957	22			
v10	Between groups	18.486	5	3.697	1.099	0.397
	Within groups	57.167	17	3.363		
	Total	75.652	22			
v11	Between groups	55.097	5	11.019	13.026	0.000
	Within groups	14.381	17	0.846		
	Total	69.478	22			
v12	Between groups	2.199	5	0.440	0.157	0.975
	Within groups	47.714	17	2.807		
	Total	49.913	22			

Appendix F. Analysis of variance between agile method groups

		Sum of squares	df	Mean square	F	Sig.
v1	Between groups	5.601	2	2.800	1.183	0.327
	Within groups	47.356	20	2.368		
	Total	52.957	22			
v2	Between groups	16.303	2	8.152	0.375	0.692
	Within groups	434.566	20	21.728		
	Total	450.870	22			
v3	Between groups	59.024	2	29.512	3.002	0.072
	Within groups	196.629	20	9.831		
	Total	255.652	22			
v4	Between groups	29.924	2	14.962	0.541	0.591
	Within groups	553.294	20	27.665		
	Total	583.217	22			
v5	Between groups	38.703	2	19.352	1.068	0.363
	Within groups	362.514	20	18.126		
	Total	401.217	22			
v6	Between groups	28.775	2	14.388	0.445	0.647
	Within groups	647.138	20	32.357		
	Total	675.913	22			
v7	Between groups	28.475	2	14.238	1.220	0.316
	Within groups	233.351	20	11.668		
	Total	261.826	22			
v8	Between groups	74.090	2	37.045	1.065	0.364
	Within groups	695.823	20	34.791		
	Total	769.913	22			
v9	Between groups	21.897	2	10.948	1.265	0.304
	Within groups	173.060	20	8.653		
	Total	194.957	22			
v10	Between groups	9.024	2	4.512	1.354	0.281
	Within groups	66.629	20	3.331		
	Total	75.652	22			
v11	Between groups	0.694	2	0.347	0.101	0.905
	Within groups	68.784	20	3.439		

Appendix F (Continued)

		Sum of squares	df	Mean square	F	Sig.
v12	Total	69.478	22			
	Between groups	6.256	2	3.128	1.433	0.262
	Within groups	43.657	20	2.183		
	Total	49.913	22			

Appendix G. Analysis of variance between organization type groups

		Sum of Squares	df	Mean Square	F	Sig.
v1	Between groups	4.367	2	2.184	0.899	0.423
	Within groups	48.589	20	2.429		
	Total	52.957	22			
v2	Between groups	29.941	2	14.970	0.711	0.503
	Within groups	420.929	20	21.046		
	Total	450.870	22			
v3	Between groups	8.420	2	4.210	0.341	0.715
	Within groups	247.232	20	12.362		
	Total	255.652	22			
v4	Between groups	56.467	2	28.234	1.072	0.361
	Within groups	526.750	20	26.338		
	Total	583.217	22			
v5	Between groups	104.360	2	52.180	3.516	0.049
	Within groups	296.857	20	14.843		
	Total	401.217	22			
v6	Between groups	262.484	2	131.242	6.349	0.007
	Within groups	413.429	20	20.671		
	Total	675.913	22			
v7	Between groups	38.898	2	19.449	1.745	0.200
	Within groups	222.929	20	11.146		
	Total	261.826	22			
v8	Between groups	278.324	2	139.162	5.662	0.011
	Within groups	491.589	20	24.579		
	Total	769.913	22			
v9	Between groups	0.582	2	0.291	0.030	0.971
	Within groups	194.375	20	9.719		
	Total	194.957	22			
v10	Between groups	20.920	2	10.460	3.882	0.039
	Within groups	54.732	20	2.737		
	Total	75.652	22			
v11	Between groups	4.675	2	2.337	0.721	0.498
	Within groups	64.804	20	3.240		
	Total	69.478	22			
v12	Between groups	8.199	2	4.099	1.965	0.166
	Within groups	41.714	20	2.086		
	Total	49.913	22			

Appendix H. Fischer's LSD post hoc method for analysis of differences among individual organizational types groups

Variable	(I) Organism	(J) Organism	Mean difference (I–J)	Std. error	Sig.
v5	Has own product	Outsourcing	0.500	1.926	0.798
		Mixed (distributed)	4.857 [*]	1.994	0.024
	Outsourcing	Has own product	–0.500	1.926	0.798
		Mixed (distributed)	4.357 [*]	1.994	0.041
	Mixed (distributed)	Has own product	–4.857 [*]	1.994	0.024
		Outsourcing	–4.357 [*]	1.994	0.041
v6	Has own product	Outsourcing	–1.000	2.273	0.665
		Mixed (distributed)	6.786 [*]	2.353	0.009
	Outsourcing	Has own product	1.000	2.273	0.665
		Mixed (distributed)	7.786 [*]	2.353	0.004
	Mixed (distributed)	Has own product	–6.786 [*]	2.353	0.009
		Outsourcing	–7.786 [*]	2.353	0.004
v8	Has own product	Outsourcing	1.125	2.479	0.655
		Mixed (distributed)	8.054 [*]	2.566	0.005
	Outsourcing	Has own product	–1.125	2.479	0.655
		Mixed (distributed)	6.929 [*]	2.566	0.014
	Mixed (distributed)	Has own product	–8.054 [*]	2.566	0.005
		Outsourcing	–6.929 [*]	2.566	0.014
v10	Has own product	Outsourcing	0.375	0.827	0.655
		Mixed (distributed)	2.232 [*]	0.856	0.017
	Outsourcing	Has own product	–0.375	0.827	0.655
		Mixed (distributed)	1.857 [*]	0.856	0.042
	Mixed (distributed)	Has own product	–2.232 [*]	0.856	0.017
		Outsourcing	–1.857 [*]	0.856	0.042

* Mean difference column represents values for which significance ≤ 0.05 .

Appendix I. Analysis of variance of answers between different team role groups

		Sum of squares	df	Mean square	F	Sig.
v1	Between groups	10.040	2	5.020	2.339	0.122
	Within groups	42.917	20	2.146		
	Total	52.957	22			
v2	Between groups	32.670	2	16.335	0.781	0.471
	Within groups	418.200	20	20.910		
	Total	450.870	22			
v3	Between groups	30.486	2	15.243	1.354	0.281
	Within groups	225.167	20	11.258		
	Total	255.652	22			
v4	Between groups	12.667	2	6.334	0.222	0.803
	Within groups	570.550	20	28.528		
	Total	583.217	22			
v5	Between groups	11.017	2	5.509	0.282	0.757
	Within groups	390.200	20	19.510		
	Total	401.217	22			
v6	Between groups	20.696	2	10.348	0.316	0.733
	Within groups	655.217	20	32.761		
	Total	675.913	22			
v7	Between groups	2.776	2	1.388	0.107	0.899
	Within groups	259.050	20	12.953		
	Total	261.826	22			
v8	Between groups	72.863	2	36.432	1.045	0.370
	Within groups	697.050	20	34.853		
	Total	769.913	22			
v9	Between groups	3.590	2	1.795	0.188	0.830
	Within groups	191.367	20	9.568		
	Total	194.957	22			
v10	Between groups	3.819	2	1.909	0.532	0.596
	Within groups	71.833	20	3.592		
	Total	75.652	22			
v11	Between groups	11.112	2	5.556	1.904	0.175
	Within groups	58.367	20	2.918		
	Total	69.478	22			
v12	Between groups	2.546	2	1.273	0.538	0.592
	Within groups	47.367	20	2.368		
	Total	49.913	22			

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